Fraudulent Claim Detection

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PROBLEM STATEMENT

Global Insure, a leading insurance company, processes thousands of claims annually. However, a significant percentage of these claims turn out to be fraudulent, resulting in considerable financial losses. The company's current process for identifying fraudulent claims involves manual inspections, which is time-consuming and inefficient. Fraudulent claims are often detected too late in the process, after the company has already paid out significant amounts. Global Insure wants to improve its fraud detection process using data-driven insights to classify claims as fraudulent or legitimate early in the approval process. This would minimise financial losses and optimise the overall claims handling process.

Business Objective

Global Insure wants to build a model to classify insurance claims as either fraudulent or legitimate based on historical claim details and customer profiles. By using features like claim amounts, customer profiles and claim types, the company aims to predict which claims are likely to be fraudulent before they are approved.

How can we analyze historical claim data to detect patterns that indicate fraudulent claims?

- We can detect patterns by performing Exploratory Data Analysis on the historical data to identify patterns in the variables involved in the claim process.
- By understanding the significance and impact of a variable using statistical measures like p-values
- By understanding the correlation between the variables and the target variables by performing :
    - *Correlation analysis*
    - *Target likelihood analysis*
    - *Boxplot analysis for Numerical variables w.r.t Target*

- Model building to predict the outcome accurately based on features with good predictive power and high accuracy

Which features are the most predictive of fraudulent behaviour?

- Based on the coefficient values for the selected variables for the logistic regression model.The most significant variable is insured_hobbies_grp_NON-ADVENTURE - BOARD GAMES which basically signifies those hobbies like Chess especially are highly prone to fraud related to claims

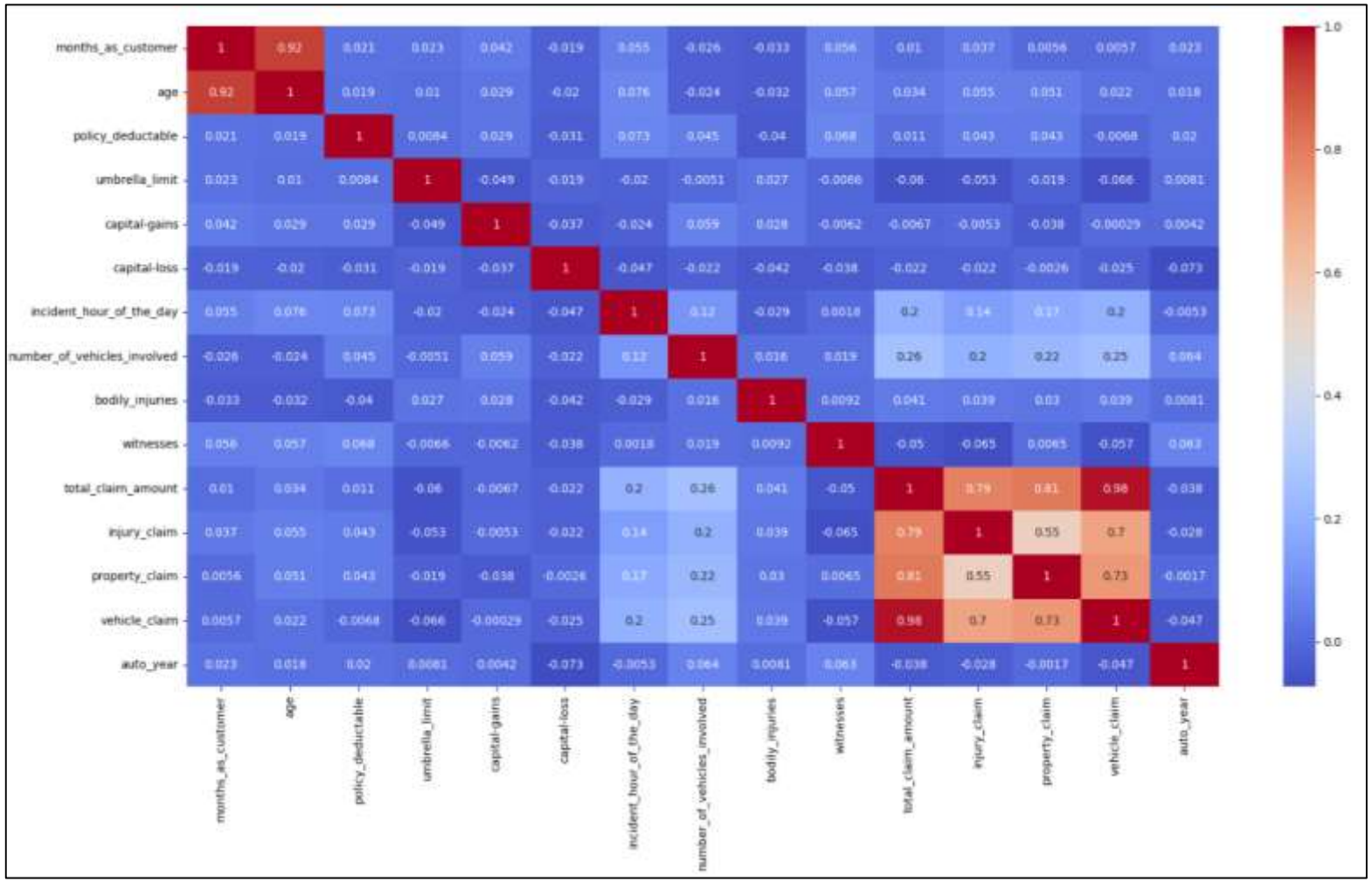
- Incident_severity_Minor Damage and incident_severity_Total Loss have a negative correlation coefficient which signifies that they are inversely proportionate to the frauds, stating that people who experience minor damages or total loss and less likely to apply for fraud claims.

BASED ON PAST DATA, CAN WE PREDICT THE LIKELIHOOD OF FRAUD FOR AN INCOMING CLAIM?

- *We built 2 models - Logistic Regression and Random Forest with decent accuracy and other metrics like Sensitivity, Specificity , Recall and Precision and based on the that, we can say, that we can predict the likelihood of Fraud for an incoming claim.*

- *The Logistic Regression model assigns a probability of fraud happening for each record highlighting the likelihood of a fraud claim whereas the Random Forest model predicts whether the claim would be fraud or not.*

Correlation Analysis



- *Age has a very high positive correlation with months_as_customer*
- *total_claim_amount has a high positive correlation with property_claim, injury_claim and vehicle claim.*

Logistic Regression – P- Values

Generalized Linear Model Regression Results

Dep. Variable:	y	No. Observations:	1052
Model:	GLM	Df Residuals:	1044
Model Family:	Binomial	Df Model:	7
Link Function:	Logit	Scale:	1.0000
Method:	IRLS	Log-Likelihood:	-457.65
Date:	Tue, 10 Jun 2025	Deviance:	915.30
Time:	16:29:47	Pearson chi2:	1.28e+03
No. Iterations:	5	Pseudo R-squ. (CS):	0.4032
Covariance Type:	nonrobust		

	coef	std err	z	P>|z|	[0.025	0.975]
const	0.0313	0.084	0.374	0.708	-0.133	0.195
incident_severity_Minor Damage	-1.6291	0.101	-16.061	0.000	-1.828	-1.430
incident_severity_Total Loss	-1.3549	0.097	-13.949	0.000	-1.545	-1.165
customer_cat_NewTenured	0.4368	0.110	3.957	0.000	0.220	0.653
customer_cat_OldTenured	0.6534	0.148	4.401	0.000	0.362	0.944
customer_cat_VeryOldTenured	0.4988	0.156	3.190	0.001	0.192	0.805
insured_hobbies_grp_NON ADVENTURE - BOARD GAMES	0.9799	0.092	10.676	0.000	0.800	1.160
insured_hobbies_grp_NON ADVENTURE - EXERCISE	0.4616	0.083	5.557	0.000	0.299	0.624

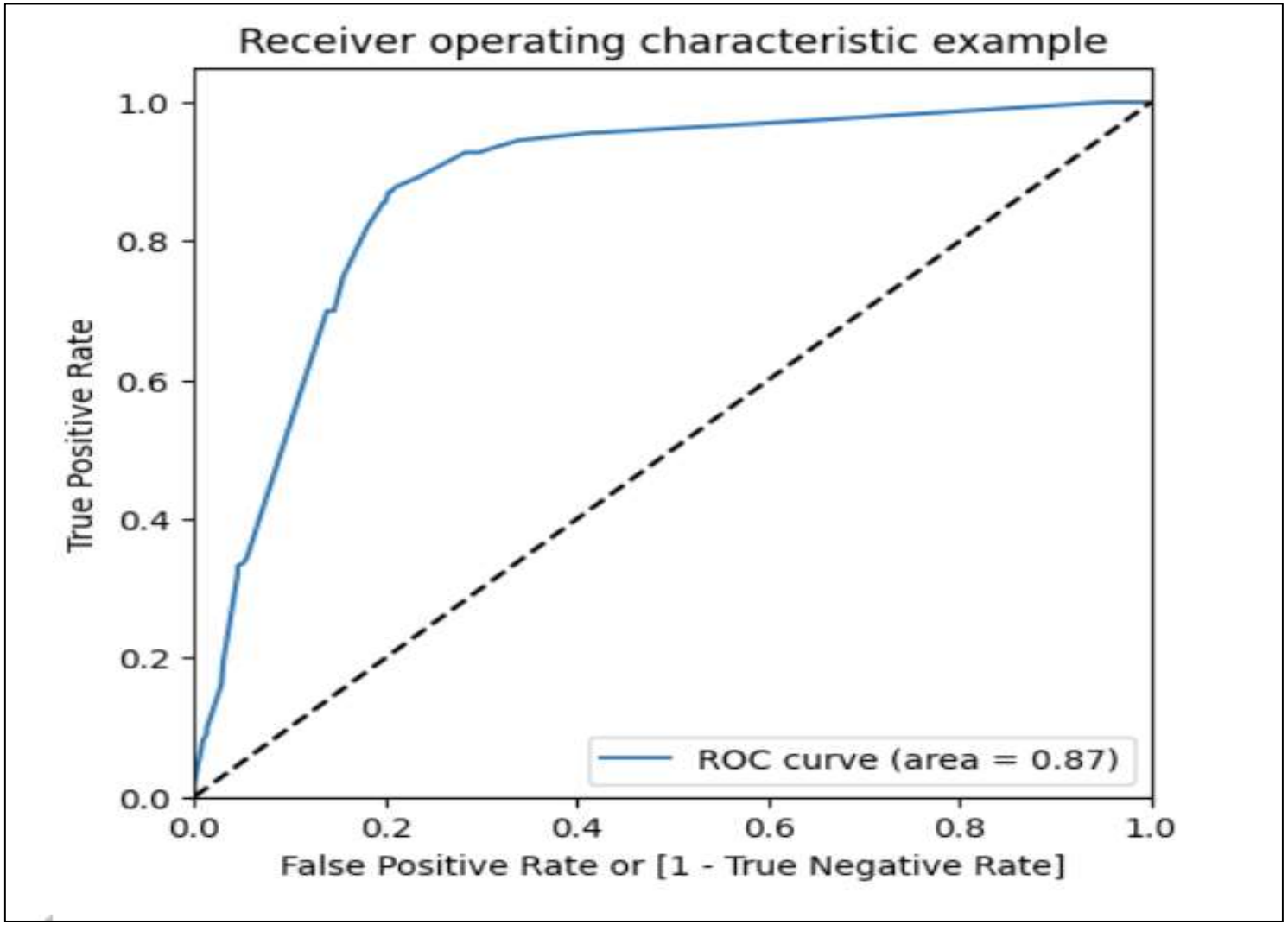
- The p-value is less than 0.05 for all the variables, stating the significance of all the selected variables.

Logistic Regression - VIF

	Features	VIF
4	customer_cat_VeryOldTenured	3.55
3	customer_cat_OldTenured	3.14
2	customer_cat_NewTenured	1.67
1	incident_severity_Total Loss	1.19
0	incident_severity_Minor Damage	1.19
5	insured_hobbies_grp_NON ADVENTURE - BOARD GAMES	1.04
6	insured_hobbies_grp_NON ADVENTURE - EXERCISE	1.03

- *The VIF values are in the acceptable range(less than 5)*

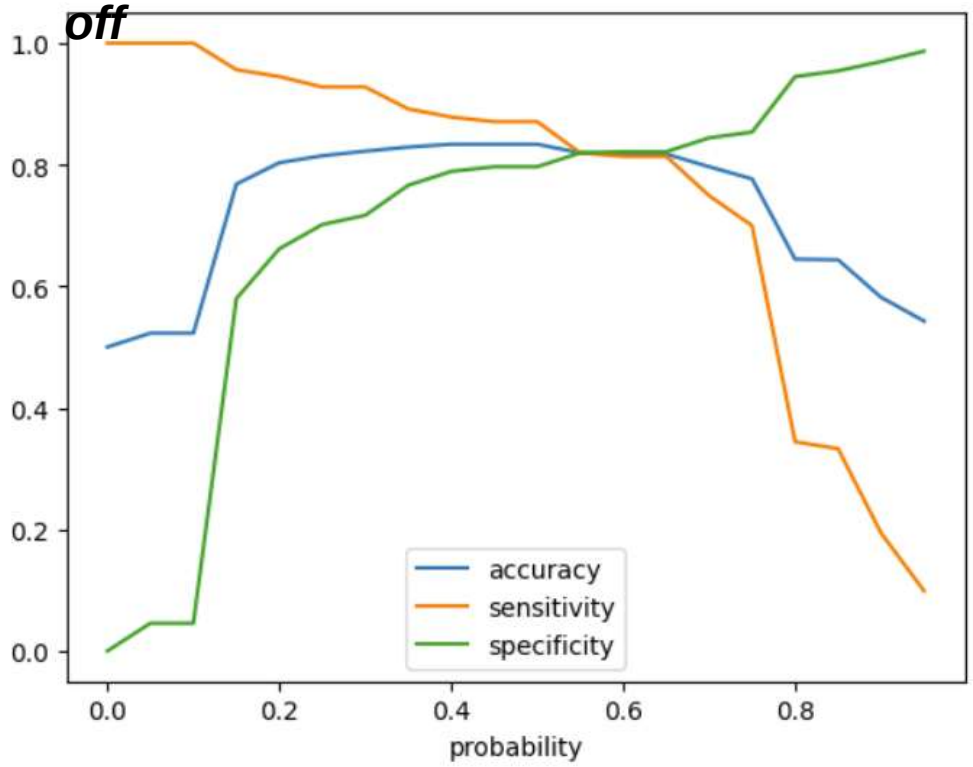
Logistic Regression - ROC Curve



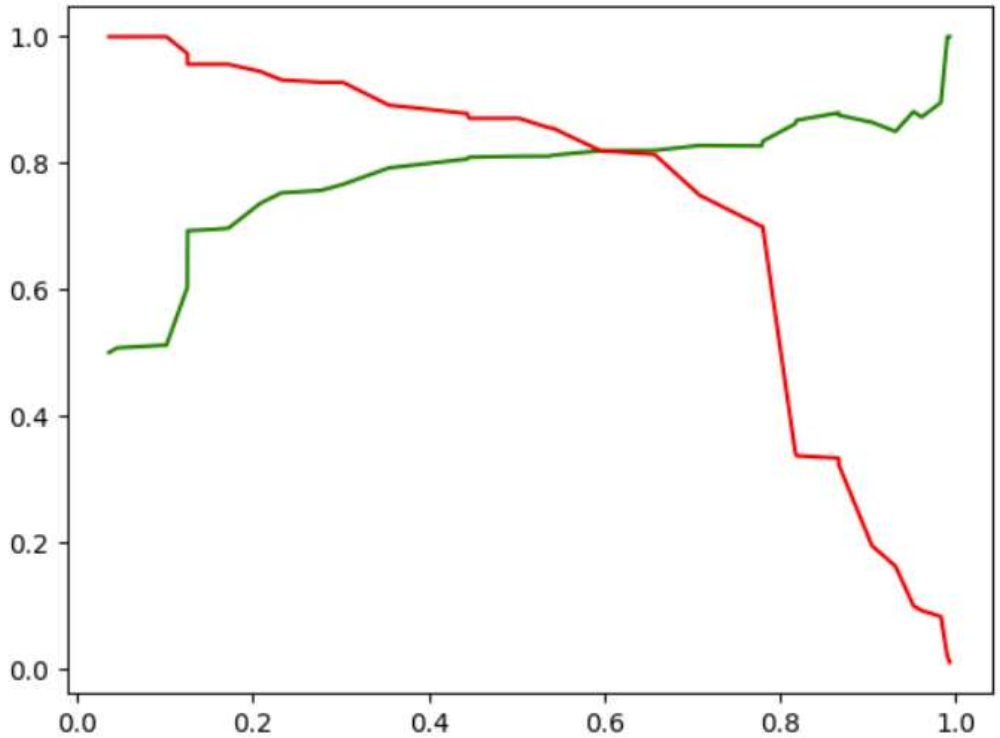
- *The AUC for the ROC curve = 0.87, which is good and states that our model is able to explain 87% of the variance.*

Logistic Regression - Optimal Cutoff

Accuracy-Sensitivity-Specificity Trade-off



Precision – Recall Trade-off



The optimal Cutoff came out as 0.6

Performance Metrics

Metric	*Logistic Regression with RFECV + statsmodels*		*Random Forest + GridSearchCV*	
	Train Set	*Validation Set*	*Train Set*	*Validation Set*
Accuracy	81.74%	80.66%	82.69%	80.66%
Recall	81.36%	74.32%	84.03%	71.62%
Precision	81.99%	58.51%	81.85%	58.88%
Specificity	82.13%	82.74%	81.36%	83.62%
Sensitivity	81.36%	74.32%	84.03%	71.62%

Insights

- Based on the accuracy scores, we can go for either of the models – logistic regression or Random Forest, but I would go with Logistic regression because of the interpretability of the model.
- Based on the coefficient values for the selected variables for the logistic regression model,
  The most significant variable is insured_hobbies_grp_NON-ADVENTURE - BOARD GAMES which basically signifies those hobbies like Chess especially are highly prone to fraud related to claims.
  incident_severity_Minor Damage and incident_severity_Total Loss have a negative correlation coefficient which signifies that they are inversely proportionate to the frauds, stating that people who experience minor damages or total loss and less likely to apply for fraud claims.
- In the Random Forest model, a different set of columns were selected based on the importances.

Recommendations

- Claims from hobbies like chess or board games should undergo additional verification
- Minor damage claims or total loss claims should not go through a rigorous process of review due to low fraud probability.
- Cases with major damage should be investigated more by the fraud detection resources.

THANK YOU